

VEHICAL-TO-GRID (V2G): How electric Vehicles can Support Renewable Energy and the Future Power Grid

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Introduction

Imagine an electric car parked outside a house not only charging from the grid, but also helping supply electricity when demand is high. This is the idea behind Vehicle-to-Grid (V2G) technology. V2G allows compatible electric vehicles (EVs) to exchange electricity with the grid through bidirectional charging. This concept is especially interesting for regions investing heavily in solar power, such as the Middle East. Solar generation is strongest during daylight, while electricity demand can remain high later in the day. EV batteries could therefore provide a flexible way to store some daytime solar electricity and return it when needed. With EV adoption also increasing across the region, V2G could connect transportation, renewable energy and electrical-grid management.

1. How Does V2G Work?

A conventional EV charger mainly moves electricity in one direction: from the grid to the vehicle. V2G adds bidirectional power flow, allowing stored energy to move back from the vehicle to the grid. A compatible EV, bidirectional charger, communication system and grid connection are required.

Vehicle to Grid Energy Flow:

Solar panels → Grid → Bidirectional charger ↔ EV battery



Home / Grid

The charging system can be controlled according to electricity demand, renewable generation and the driver's needs. For example, an EV could charge when solar generation is high and later discharge a controlled amount during an evening demand peak. The vehicle would retain a minimum charge so that it remains useful to its owner.

The technology is still developing. The International Energy Agency reported in 2026 that only 22 EV models had commercial V2G capability, representing less than 1.5% of EV models. This shows that V2G is promising but has not yet become a mainstream feature.

2. Why the Middle East Could Benefit

The Middle East has strong potential for solar power, making it an interesting region for V2G development. IRENA reports that solar PV was the largest source of new renewable capacity globally in 2024, adding more than 450 GW. In 2025, solar PV additions in Saudi Arabia nearly quadrupled to almost 7 GW, according to the IEA. At the same time, electric-car sales in the Middle East reached about 75,000 in 2025, increasing by more than 40% in one year. The UAE accounted for almost half of those regional sales, while Saudi Arabia and Qatar together represented nearly 45%.

Dataset:

Indicator	Value	Year
Global solar PV additions	452.1 GW	2024
Saudi Arabia solar PV additions	~7 GW	2025
Middle East electric-car sales	~75,000	2025
Middle East EV sales growth	>40%	2025

These figures do not mean V2G is already widely available in the region. Instead, they show why the region could be a useful environment for future pilot programmes: solar capacity is expanding while the EV market is growing.

V2G could help manage the mismatch between solar production and electricity demand. During sunny hours, EVs could absorb surplus solar electricity. Later, when demand rises, selected vehicles could return part of that energy to the grid. This could reduce peak loads and improve the usefulness of renewable generation.

For the region, timing would matter. A solar-linked charging strategy could prioritise charging during periods of high photovoltaic output rather than charging whenever a vehicle is plugged in. This would make the EV fleet more responsive to generation. In cities with many parked vehicles, an aggregator could combine many cars and communicate with the utility. The utility would not need to control cars individually; instead, it could request some flexibility, while the aggregator distributes that request among participating vehicles. Such a system could also create a new role for EV owners, who could receive payments or lower charging costs in exchange for making battery capacity available and reduce peak demand.

3. Challenges and Possible Solutions

Several barriers must be solved before V2G can scale. The first is compatibility. Different EVs and chargers may use different systems, making communication difficult. A practical solution would be regional standards based on established international communication protocols, followed by certification tests for vehicles and chargers.

The second challenge is battery degradation. Frequent charging and discharging could increase battery wear if poorly managed. Smart software could solve part of this problem by setting charging limits, avoiding unnecessary cycling and only using vehicles that have enough available capacity. Owners should also be allowed to set minimum battery levels and departure times.

Cost is another barrier. The IEA reports that bidirectional AC chargers can cost below USD 1,500, while DC options can cost USD 5,000 or more. Governments and utilities could reduce the initial barrier through pilot subsidies, time-of-use electricity tariffs and payments to EV owners who provide grid services.

A final solution is to begin with controlled pilot projects rather than attempting immediate large-scale deployment. Solar-powered charging hubs at offices, universities, residential communities or public parking areas could test V2G under local conditions. Aggregators could coordinate many participating vehicles as one flexible energy resource.

Conclusion

V2G should not be viewed as a replacement for large-scale energy storage. Instead, it can complement solar farms, batteries and smarter electricity networks. The Middle East's growing solar capacity and expanding EV market create a strong reason to investigate this technology, although regional standards, costs and battery-management concerns must be addressed first. Carefully designed pilot projects could provide the data needed to determine where V2G creates the greatest value. If implemented responsibly, tomorrow's electric cars could become more than a form of transportation: they could also become part of the region's renewable-energy infrastructure.

Citations

- [IEA — Vehicle-to-Grid Technology \(2026\)](#) — V2G models, charger costs, standards, benefits and challenges.
- [IEA — Global EV Outlook 2026](#) — Middle East EV sales data.
- [IRENA — Renewable Capacity Statistics 2026](#) — renewable-energy dataset.
- [IRENA — Solar Energy](#) — global solar PV statistics.